

PRATYUSHA MALLELA

(971) 895-4741 — prathyushamallela3@gmail.com — linkedin.com/in/prathyusha-mallela
website: prathyushamallela.github.io — github: github.com/prathyushamallela

SUMMARY

- Software Engineer with **8+ years of industry experience** building **backend, distributed, and ML-driven systems** in fintech and data-intensive environments. **Work Authorization:** OPT EAD – Authorized for immediate U.S. employment.
- Recently graduated with an **M.S. in Computer Science**, focusing on **Artificial Intelligence**, from Oregon State University (Corvallis, OR).
- Strong proficiency in **Java and Python**, with deep **object-oriented design** expertise and experience architecting systems from first principles.
- Hands-on experience across the **entire Machine Learning lifecycle**, including ETL, feature engineering, model training, evaluation, deployment, and monitoring.
- Built and deployed **production ML models** using **PyTorch** for fraud detection and automated decisioning.
- Experience with **event-driven and distributed systems** using **Kafka, Spark, Scala-based pipelines**, and **ZooKeeper**.
- **AWS Certified Cloud Practitioner** — [Credly Public Badge](#)
- Master's projects in AI (OSU):
 - **Causal-temporal modeling** for supply-chain systems using inference-based techniques. (using Causal Inferencing and Probabilistic Graphical Models)
 - **Hybrid rule-based NLP** with **text-to-text transformer models (T5)** for solving cryptic crosswords.
- Designed and maintained **Pentaho (Kettle) ETL pipelines** for large-scale transactional ingestion and ML feature pipelines.
- Familiar with **LangChain and LangGraph** for LLM-driven workflows and AI orchestration.
- **Code, research prototypes, and system designs** available at: github.com/prathyushamallela

EDUCATION

M.S. in Computer Science, focus in AI Oregon State University, USA *Mar 2025* — GPA: 3.67

TECHNICAL SKILLS

Languages: Java (Core/EE), Python

Backend & Architecture: Spring Boot, Spring MVC, Hibernate, Drools, REST APIs, SOAP, Microservices, POJO Design, AWS

ML/AI: PyTorch, CNNs, Kohonen SOMs, Feature Engineering, Model Training, Deployment, Experimentation, LLM, LangChain, LangGraph

Distributed Systems: Apache Kafka, Apache Spark, Scala, ZooKeeper, Distributed Coordination

ETL / Data Engineering: Pentaho (Kettle), BigQuery

Databases: Oracle, IBM Db2, MSSQL, MongoDB, SOQL, JDBC

DevOps / Tooling: GitHub, Docker, Maven, JUnit

PROFESSIONAL EXPERIENCE

Software Engineer – Backend & ML Systems
Manipal Technologies Ltd

2016 – 2020

- **AI-driven Fraud Detection Solutions for international banks**

- Developed ubiquitous fraud classification models to detect suspicious bank transactions for multiple international banks, preventing multimillion-dollar losses.
- Achieved **83% false-positive reduction**, **80% fraud capture**, and **40% reduction in manual review effort**, saving over **\$2M annually**.
- Reduced fraud false positives using **Kohonen Self-Organizing Maps (SOMs)** integrated with **Drools rule engines**.
- Designed an **11-layer CNN-based fraud detection model** achieving **80% fraud capture**.
- Developed a **case-worker recommendation system** to prioritize high-risk transactions and improve investigation efficiency.
- Applied **data augmentation** techniques to generate synthetic data of fraud rules and investigation scenarios.
- Feature Engineering: Built **near real-time transaction scoring pipelines** using **Apache Kafka** and **Pentaho (Kettle) ETL** for feature generation.
- Designed **enterprise-grade backend services** using **Java**, **Spring Boot**, **Spring MVC**, **Hibernate**, integrating ML models with **rule-based decision engines**.
- Implemented a modular **distributed fraud platform** leveraging **Kafka** for streaming, **ZooKeeper** for coordination, and **Spark/Scala** for large-scale analytics.

- **Anti-Money Laundering (AML) solutions**

- Led migration from rule-only systems to **hybrid ML-driven architectures**.
- Designed **near real-time AML pipelines** using **Kafka**, **Apache Spark**, and **Pentaho ETL**.

Product Developer
MVR Info Systems

2012 – 2016

- Built backend services for a Smart Energy Alert System using REST APIs and relational databases.
- Developed Angular-based UI workflows for real-time energy monitoring.

Salesforce Programmer Analyst
Oregon State University

Jul 2022 – Oct 2023

- Supported and enhanced **Salesforce CRM-based university applications**, including the **Graduate Committee Representative List UI** and backend services.
- Performed backend testing and issue triage for the **Exam Scheduling Form**, ensuring data validation, workflow automation, and system reliability.
- Conducted **UAT**, **regression testing**, and **workflow validation** within Agile sprint cycles.
- Diagnosed and resolved **production issues**, maintaining application stability, data integrity, and reliable user workflows.